

ek_counterfactuals

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2026-09-22

Libraries

```
library(here)
library(tidyverse)
```

Question 1

Write down the model

$$\sum_i X_{ni}^0 = \text{total spending of } n \quad (1)$$

$$\sum_d X_{di}^0 = \text{total sales, hence income, of } i = w_i^0 L_i^0 \quad (2)$$

$$F_i(z) = \exp(-T_i z^{-\theta}) \quad (3)$$

$$\Phi_n = \sum_{o=1}^N T_o (w_o d_{no})^{-\theta}, \quad P_n = \gamma \Phi_n^{-1/\theta} \quad (4)$$

$$\pi_{ni} = \frac{T_i (w_i d_{ni})^{-\theta}}{\Phi_n} \quad (5)$$

$$w_i L_i = \sum_{n=1}^N \pi_{ni} X_n \quad (6)$$

$$X_n = w_n L_n + h(w) D_n, \quad \sum_{n=1}^N D_n = 0 \quad (7)$$

$$\pi'_{ni} = \frac{\hat{T}_i T_i^0 \left(\hat{w}_i w_i^0 \hat{d}_{ni} d_{ni}^0 \right)^{-\theta}}{\sum_o \hat{T}_o T_o^0 \left(\hat{w}_o w_o^0 \hat{d}_{no} d_{no}^0 \right)^{-\theta}} \quad (8)$$

$$\underbrace{\hat{T}_i \left(\hat{w}_i \hat{d}_{ni} \right)^{-\theta}}_{\text{hats}} \cdot \underbrace{T_i^0 \left(w_i^0 d_{ni}^0 \right)^{-\theta}}_{= \Phi_n^0 \pi_{ni}^0} \quad (9)$$

$$\hat{\pi}_{ni} = \frac{\pi'_{ni}}{\pi^0_{ni}} = \frac{\hat{T}_i \left(\hat{d}_{ni} \hat{w}_i \right)^{-\theta}}{\sum_{o=1}^N \pi^0_{no} \hat{T}_o \left(\hat{d}_{no} \hat{w}_o \right)^{-\theta}} \quad (10)$$

$$\hat{\Phi}_n = \sum_{o=1}^N \pi^0_{no} \hat{T}_o \left(\hat{d}_{no} \hat{w}_o \right)^{-\theta}, \quad \hat{P}_n = \hat{\Phi}_n^{-1/\theta} \quad (11)$$

$$y_n^0 \equiv \frac{w_n^0 L_n^0}{X_n^0} = \frac{\sum_i X_{in}^0}{\sum_i X_{ni}^0} \quad (12)$$

$$h(w^0) D_n = (1 - y_n^0) X_n^0 \quad (13)$$

$$w'_n L'_n = \hat{w}_n \hat{L}_n w_n^0 L_n^0 = \hat{w}_n \hat{L}_n y_n^0 X_n^0 \quad (14)$$

$$h(w') D'_n = h(w^0 \hat{w}) \hat{D}_n D_n = \frac{h(w^0 \hat{w})}{h(w^0)} \hat{D}_n \cdot h(w^0) D_n = \frac{h(w^0 \hat{w})}{h(w^0)} \hat{D}_n (1 - y_n^0) X_n^0 \quad (15)$$

$$\frac{X'_n}{X_n^0} = y_n^0 \hat{w}_n \hat{L}_n + (1 - y_n^0) \frac{h(w^0 \hat{w})}{h(w^0)} \hat{D}_n \quad (16)$$

$$\hat{w}_i \hat{L}_i = \sum_{n=1}^N \frac{\pi'_{ni} X'_n}{Y_i^0} \quad (17)$$

$$\frac{\pi'_{ni} X'_n}{Y_i^0} = \underbrace{\frac{\pi'_{ni}}{\pi^0_{ni}}}_{\text{Step 1}} \cdot \underbrace{\frac{\pi^0_{ni} X_n^0}{Y_i^0}}_{= X_{ni}^0 / \sum_d X_{di}^0} \cdot \underbrace{\frac{X'_n}{X_n^0}}_{\text{Step 2}} \quad (18)$$

$$Z_i(\hat{w}) \equiv \sum_{n=1}^N \frac{X_{ni}^0}{\sum_{d=1}^N X_{di}^0} \cdot \frac{\hat{T}_i \left(\hat{d}_{ni} \hat{w}_i \right)^{-\theta}}{\sum_o \pi^0_{no} \hat{T}_o \left(\hat{d}_{no} \hat{w}_o \right)^{-\theta}} \left(y_n^0 \hat{w}_n \hat{L}_n + (1 - y_n^0) \frac{h(w^0 \hat{w})}{h(w^0)} \hat{D}_n \right) - \hat{w}_i \hat{L}_i \quad (19)$$

$$Z_i(\hat{w}) = 0 \quad \text{for all } i \quad (20)$$

Question 2

The trade data give the matrix of bilateral flows X_{ni}^0 directly, with `country_dest` indexing the buyer n and `country_org` the seller i . Row sums are total spending and column sums are total income, from which I construct the expenditure shares π_{ni}^0 , the income-to-spending ratios y_n^0 , and the share of each country's sales going to each destination. The trade elasticity is set to $\theta = 4$.

#download the data

```
country_list <- read_csv(here("data", "country_list.csv"))
trade <- read_csv(here("data", "bilateral_trade_country.csv"))
```

#set variable names

```
countries <- country_list$country
```

```

N      <- length(countries)

# trade elasticity, theta set to 4 as prescribed
theta <- 4

# building the matrix X0[n, i] = spending by n (destination) on goods from i (origin)
X0 <- matrix(0, nrow = N, ncol = N,
             dimnames = list(countries, countries))
X0[cbind(trade$country_dest, trade$country_org)] <- trade$trade_flow2014

# total spending of n = row sums; total income of i = column sums
spending <- rowSums(X0)
income   <- colSums(X0)

# expenditure shares pi0[n, i]
pi0 <- X0 / spending

# income-to-spending ratios
y0 <- income / spending

# share of i's sales going to n
sales_share <- t(t(X0) / income)

```

Question 3

The shock is a 10% rise in Chinese productivity, $\hat{T}_{\text{China}} = 1.1$, with trade costs, transfers and labour forces held fixed. Transfers are denominated in US wages and the US wage is the numeraire, so $h(w^0 \hat{w})/h(w^0) = 1$ and the transfer term in Z_i is inert. I solve the system by iterating on excess demand until the largest $|Z_i|$ falls below 10^{-10} .

(a) Initial guess

```

# shocks: 10% rise in Chinese productivity, nothing else
That <- rep(1, N)
names(That) <- countries
That["CHN"] <- 1.1

Dhat <- rep(1, N)
Lhat <- rep(1, N)

# h(w) = w_US and what_US = 1, so the numeraire ratio is one
h_ratio <- 1

# step b = 0: guess what_i = 1 for all i
what <- rep(1, N)
names(what) <- countries

# US wage is the numeraire, fixed at 1 throughout
what["USA"] <- 1

what

## AUS AUT BEL BGR BRA CAN CHE CHN CZE DEU DNK ESP EST FIN FRA GBR GRC HRV HUN IDN
##   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1   1

```

```
## IND IRL ITA JPN KOR LTU LVA MEX NLD NOR POL PRT ROU ROW RUS SVK SVN SWE TUR TWN
## 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
## USA
## 1
```

(b) Iteration

```
kappa <- 0.5
tol <- 1e-10
B <- 100000

# store the error at each step
err_path <- rep(NA, B)

what <- rep(1, N)
names(what) <- countries

for (b in 1:B) {

  Phihat <- as.vector(pi0 %*% (That * what^(-theta)))
  pihat <- matrix(That * what^(-theta), nrow = N, ncol = N, byrow = TRUE) / Phihat
  Xhat <- y0 * what * Lhat + (1 - y0) * h_ratio * Dhat

  Z <- colSums(sales_share * pihat * Xhat) - what * Lhat
  Z["USA"] <- 0

  err_path[b] <- max(abs(Z))

  if (err_path[b] < tol) break

  what <- what + kappa * Z
  what["USA"] <- 1
}

err_path <- err_path[1:b]

b
```

```
## [1] 258
```

```
err_path[b]
```

```
## [1] 9.963497e-11
```

Convergence check

The maximum error falls monotonically at every step, from 1.2×10^{-2} to below tolerance in 258 iterations. On a log scale the decline is close to linear, which is what a well-chosen κ should produce.

```
# error falls monotonically at every step
all(diff(err_path) < 0)
```

```
## [1] TRUE
```

```
# first few and last few errors
head(err_path)
```

```
## [1] 0.012416770 0.007537234 0.004942981 0.003368680 0.002370837 0.001720655
```

```
tail(err_path)
```

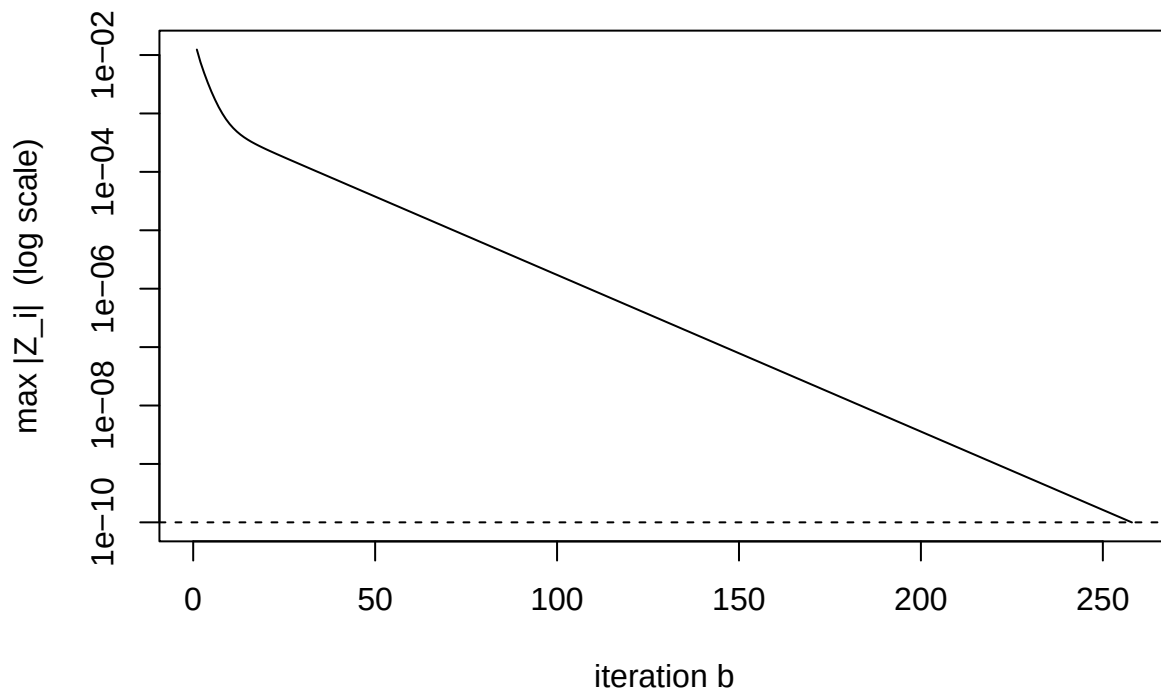
```
## [1] 1.356797e-10 1.275533e-10 1.199154e-10 1.127336e-10 1.059826e-10
```

```
## [6] 9.963497e-11
```

```
# log scale makes the geometric decline visible
```

```
plot(err_path, type = "l", log = "y",
     xlab = "iteration b",
     ylab = "max |Z_i| (log scale)",
     main = "Convergence of the excess demand error")
abline(h = tol, lty = 2)
```

Convergence of the excess demand error



Question 4

Real wage and real expenditure

```
# change in market access and price index
Phihat <- as.vector(pi0 %*% (That * what^(-theta)))
Phat   <- Phihat^(-1 / theta)
names(Phihat) <- countries
names(Phat)   <- countries

# change in total spending
Xhat <- y0 * what * Lhat + (1 - y0) * h_ratio * Dhat
```

```

# change in domestic trade share
pihat <- matrix(That * what^(-theta), nrow = N, ncol = N, byrow = TRUE) / Phihat
dimnames(pihat) <- list(countries, countries)
pihat_ii <- diag(pihat)

# initial spending share on Chinese goods
pi0_china <- pi0[, "CHN"]

results <- data.frame(
  country   = countries,
  what      = what,
  Xhat      = Xhat,
  Phat      = Phat,
  real_wage = what / Phat,
  real_exp  = Xhat / Phat,
  pihat_ii  = pihat_ii,
  Phihat    = Phihat,
  pi0_china = pi0_china
)

results[order(-results$real_wage), ]

```

##	country	what	Xhat	Phat	real_wage	real_exp	pihat_ii
##	CHN	CHN 1.0217721	1.0221792	0.9978595	1.023964	1.024372	1.0005851
##	TWN	TWN 1.0005778	1.0006164	1.0003787	1.000199	1.000238	0.9992044
##	KOR	KOR 1.0003699	1.0003852	1.0002556	1.000114	1.000130	0.9995428
##	ROW	ROW 1.0001650	1.0001572	1.0000628	1.000102	1.000094	0.9995916
##	AUS	AUS 1.0003551	1.0003549	1.0002894	1.000066	1.000066	0.9997370
##	NLD	NLD 1.0000749	1.0000809	1.0000205	1.000054	1.000060	0.9997823
##	JPN	JPN 1.0001509	1.0001497	1.0001042	1.000047	1.000045	0.9998132
##	DEU	DEU 1.0001611	1.0001705	1.0001177	1.000043	1.000053	0.9998262
##	IDN	IDN 1.0001164	1.0001169	1.0000732	1.000043	1.000044	0.9998274
##	RUS	RUS 1.0000630	1.0000653	1.0000230	1.000040	1.000042	0.9998397
##	HUN	HUN 1.0001027	1.0001069	1.0000648	1.000038	1.000042	0.9998483
##	CZE	CZE 1.0000889	1.0000926	1.0000550	1.000034	1.000038	0.9998643
##	FIN	FIN 1.0001022	1.0001028	1.0000710	1.000031	1.000032	0.9998752
##	DNK	DNK 1.0001400	1.0001464	1.0001095	1.000031	1.000037	0.9998780
##	EST	EST 1.0000547	1.0000549	1.0000252	1.000030	1.000030	0.9998818
##	IRL	IRL 1.0001101	1.0001225	1.0000808	1.000029	1.000042	0.9998825
##	CAN	CAN 1.0000035	1.0000035	0.9999764	1.000027	1.000027	0.9998916
##	BRA	BRA 1.0001722	1.0001702	1.0001465	1.000026	1.000024	0.9998973
##	SWE	SWE 1.0001022	1.0001055	1.0000766	1.000026	1.000029	0.9998975
##	MEX	MEX 0.9999688	0.9999687	0.9999443	1.000024	1.000024	0.9999020
##	SVK	SVK 1.0001402	1.0001447	1.0001163	1.000024	1.000028	0.9999045
##	BEL	BEL 1.0001025	1.0001060	1.0000788	1.000024	1.000027	0.9999051
##	TUR	TUR 1.0000297	1.0000301	1.0000078	1.000022	1.000022	0.9999123
##	SVN	SVN 1.0000778	1.0000796	1.0000574	1.000020	1.000022	0.9999183
##	POL	POL 1.0000681	1.0000693	1.0000483	1.000020	1.000021	0.9999210
##	GBR	GBR 1.0000803	1.0000800	1.0000614	1.000019	1.000019	0.9999243
##	FRA	FRA 1.0001018	1.0001014	1.0000834	1.000018	1.000018	0.9999267
##	IND	IND 1.0000598	1.0000598	1.0000415	1.000018	1.000018	0.9999268
##	USA	USA 1.0000000	1.0000000	0.9999818	1.000018	1.000018	0.9999273
##	BGR	BGR 1.0001019	1.0000996	1.0000847	1.000017	1.000015	0.9999310
##	CHE	CHE 1.0001289	1.0001375	1.0001133	1.000016	1.000024	0.9999376

##	AUT	AUT	1.0001388	1.0001417	1.0001234	1.000015	1.000018	0.9999382
##	ITA	ITA	1.0000956	1.0000973	1.0000812	1.000014	1.000016	0.9999421
##	NOR	NOR	1.0001148	1.0001240	1.0001005	1.000014	1.000023	0.9999430
##	LVA	LVA	1.0000548	1.0000541	1.0000415	1.000013	1.000013	0.9999470
##	ESP	ESP	1.0000675	1.0000677	1.0000553	1.000012	1.000012	0.9999508
##	LTU	LTU	1.0000684	1.0000703	1.0000564	1.000012	1.000014	0.9999518
##	GRC	GRC	1.0000486	1.0000464	1.0000385	1.000010	1.000008	0.9999594
##	ROU	ROU	1.0000829	1.0000830	1.0000736	1.000009	1.000009	0.9999628
##	HRV	HRV	1.0000773	1.0000776	1.0000693	1.000008	1.000008	0.9999680
##	PRT	PRT	1.0000832	1.0000819	1.0000753	1.000008	1.000007	0.9999687
##		Phihat	pi0_china					
##	CHN		1.0086081					0.940864195
##	TWN		0.9984866					0.038093336
##	KOR		0.9989784					0.031167650
##	ROW		0.9997486					0.038524100
##	AUS		0.9988434					0.017770540
##	NLD		0.9999179					0.027556913
##	JPN		0.9995832					0.019774488
##	DEU		0.9995294					0.013242480
##	IDN		0.9997072					0.020488797
##	RUS		0.9999081					0.019961525
##	HUN		0.9997408					0.019728520
##	CZE		0.9997800					0.018795229
##	FIN		0.9997162					0.013454298
##	DNK		0.9995620					0.010541882
##	EST		0.9998993					0.019743926
##	IRL		0.9996770					0.007547396
##	CAN		1.0000944					0.015331277
##	BRA		0.9994144					0.009393282
##	SWE		0.9996936					0.011324230
##	MEX		1.0002227					0.018011043
##	SVK		0.9995350					0.008982024
##	BEL		0.9996848					0.009937147
##	TUR		0.9999688					0.015695445
##	SVN		0.9997705					0.014693565
##	POL		0.9998066					0.013178075
##	GBR		0.9997546					0.009779444
##	FRA		0.9996663					0.008197795
##	IND		0.9998340					0.011255507
##	USA		1.0000727					0.011039487
##	BGR		0.9996615					0.008185023
##	CHE		0.9995468					0.005547591
##	AUT		0.9995066					0.005346894
##	ITA		0.9996754					0.007204193
##	NOR		0.9995979					0.005885384
##	LVA		0.9998339					0.009941231
##	ESP		0.9997790					0.008393259
##	LTU		0.9997745					0.011284320
##	GRC		0.9998461					0.010651179
##	ROU		0.9997056					0.006568035
##	HRV		0.9997226					0.007353824
##	PRT		0.9996987					0.005348431

China's real wage rises 2.4%. Every other country gains, but by less than 0.02%. Mexico is the only country

whose nominal wage falls, since it competes directly with China in the US market; its real wage still rises because its price index falls further.

Regressions

```
# drop China
reg_data <- subset(results, country != "CHN")

reg_data$ln_real_wage <- log(reg_data$real_wage)
reg_data$ln_real_exp <- log(reg_data$real_exp)
reg_data$ln_pihat_ii <- log(reg_data$pihat_ii)
reg_data$ln_Phihat <- log(reg_data$Phihat)

# real wage on each regressor separately
summary(lm(ln_real_wage ~ ln_pihat_ii, data = reg_data))

##
## Call:
## lm(formula = ln_real_wage ~ ln_pihat_ii, data = reg_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.454e-16 -6.033e-17  1.974e-18  5.187e-17  1.507e-16
##
## Coefficients:
##              Estimate Std. Error    t value Pr(>|t|)
## (Intercept)  1.831e-17  1.829e-17  1.001e+00   0.323
## ln_pihat_ii -2.500e-01  9.506e-14 -2.630e+12  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.315e-17 on 38 degrees of freedom
## Multiple R-squared:  1, Adjusted R-squared:  1
## F-statistic: 6.916e+24 on 1 and 38 DF, p-value: < 2.2e-16
summary(lm(ln_real_wage ~ ln_Phihat, data = reg_data))

##
## Call:
## lm(formula = ln_real_wage ~ ln_Phihat, data = reg_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.263e-05 -1.756e-05 -5.850e-06  1.175e-05  7.455e-05
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.924e-06  5.838e-06   1.357   0.183
## ln_Phihat   -7.816e-02  1.299e-02  -6.019 5.36e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.538e-05 on 38 degrees of freedom
## Multiple R-squared:  0.4881, Adjusted R-squared:  0.4746
## F-statistic: 36.23 on 1 and 38 DF, p-value: 5.358e-07
```

```
summary(lm(ln_real_wage ~ pi0_china, data = reg_data))

##
## Call:
## lm(formula = ln_real_wage ~ pi0_china, data = reg_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.819e-05 -1.164e-05  1.100e-08  9.331e-06  7.780e-05
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.825e-05  5.814e-06  -3.138  0.00328 **
## pi0_china    3.660e-03  3.572e-04  10.247  1.73e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.829e-05 on 38 degrees of freedom
## Multiple R-squared:  0.7343, Adjusted R-squared:  0.7273
## F-statistic: 105 on 1 and 38 DF, p-value: 1.727e-12
# real expenditure on each regressor separately
summary(lm(ln_real_exp ~ ln_pihat_ii, data = reg_data))

##
## Call:
## lm(formula = ln_real_exp ~ ln_pihat_ii, data = reg_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.055e-05 -1.853e-06  7.280e-08  1.500e-06  1.203e-05
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.764e-06  1.154e-06  -1.528   0.135
## ln_pihat_ii -2.856e-01  6.000e-03 -47.606 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.248e-06 on 38 degrees of freedom
## Multiple R-squared:  0.9835, Adjusted R-squared:  0.9831
## F-statistic: 2266 on 1 and 38 DF, p-value: < 2.2e-16
summary(lm(ln_real_exp ~ ln_Phihat, data = reg_data))

##
## Call:
## lm(formula = ln_real_exp ~ ln_Phihat, data = reg_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4.853e-05 -2.006e-05 -5.411e-06  1.227e-05  9.017e-05
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)  5.946e-06  6.476e-06   0.918   0.364
## ln_Phihat   -9.343e-02  1.441e-02  -6.485  1.23e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.816e-05 on 38 degrees of freedom
## Multiple R-squared:  0.5253, Adjusted R-squared:  0.5128
## F-statistic: 42.05 on 1 and 38 DF,  p-value: 1.234e-07
summary(lm(ln_real_exp ~ pi0_china, data = reg_data))

##
## Call:
## lm(formula = ln_real_exp ~ pi0_china, data = reg_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.904e-05 -1.517e-05 -1.245e-06  8.620e-06  1.059e-04
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -1.967e-05  7.665e-06  -2.567   0.0143 *
## pi0_china    3.974e-03  4.709e-04   8.439 3.06e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.411e-05 on 38 degrees of freedom
## Multiple R-squared:  0.6521, Adjusted R-squared:  0.6429
## F-statistic: 71.21 on 1 and 38 DF,  p-value: 3.055e-10
```

The regression of the log real wage on $\ln \hat{\pi}_{ii}$ returns a coefficient of exactly -0.25 with $R^2 = 1$. This is not an empirical finding but an identity: for every country other than China $\hat{T}_i = 1$, so $\hat{\pi}_{ii} = \hat{w}_i^{-\theta} / \hat{\Phi}_i$ and $\hat{\pi}_{ii}^{-1/\theta} = \hat{w}_i / \hat{\Phi}_i$. The estimate recovers $-1/\theta$ because the model says it must. I take the exact fit as a check that the solver is coded correctly.

Market access enters with a negative coefficient, which looks wrong until one notices that $\hat{\Phi}_i$ is endogenous. Each country carries a large weight in its own Φ , so a country whose wage rises sees its own $\hat{\Phi}$ fall. Taiwan has both the largest real wage gain and the lowest $\hat{\Phi}$. The partial effect of market access is positive, but the regression picks up the correlation with $\hat{w}_i$ and reverses the sign.

The initial Chinese import share gives the cleanest result: a coefficient of 0.0037 with an R^2 of 0.73. Countries that bought more from China to begin with gain more, because the productivity improvement reaches them through cheaper imports in proportion to how much of their spending goes to Chinese goods. The real expenditure regressions behave the same way, except that the coefficient on $\ln \hat{\pi}_{ii}$ is -0.286 rather than -0.25 . The ACR identity holds for real income, not real expenditure, and the gap between them is the transfer.

Question 5

Solve with the Chinese wage as the transfer numeraire

```
kappa <- 0.5
tol    <- 1e-10
B      <- 100000

# store the error at each step
err_path2 <- rep(NA, B)
```

```

what2 <- rep(1, N)
names(what2) <- countries

for (b in 1:B) {

  Phihat2 <- as.vector(pi0 %*% (That * what2^(-theta)))
  pihat2 <- matrix(That * what2^(-theta), nrow = N, ncol = N, byrow = TRUE) / Phihat2

  #  $h(w_0 \text{ what}) / h(w_0) = \text{what\_China}$ 
  h_ratio2 <- what2["CHN"]

  Xhat2 <- y0 * what2 * Lhat + (1 - y0) * h_ratio2 * Dhat

  Z2 <- colSums(sales_share * pihat2 * Xhat2) - what2 * Lhat
  Z2["USA"] <- 0

  err_path2[b] <- max(abs(Z2))

  if (err_path2[b] < tol) break

  what2 <- what2 + kappa * Z2
  what2["USA"] <- 1
}

err_path2 <- err_path2[1:b]

b

## [1] 247
err_path2[b]

## [1] 9.779999e-11

Convergence check

all(diff(err_path2) < 0)

## [1] TRUE
head(err_path2)

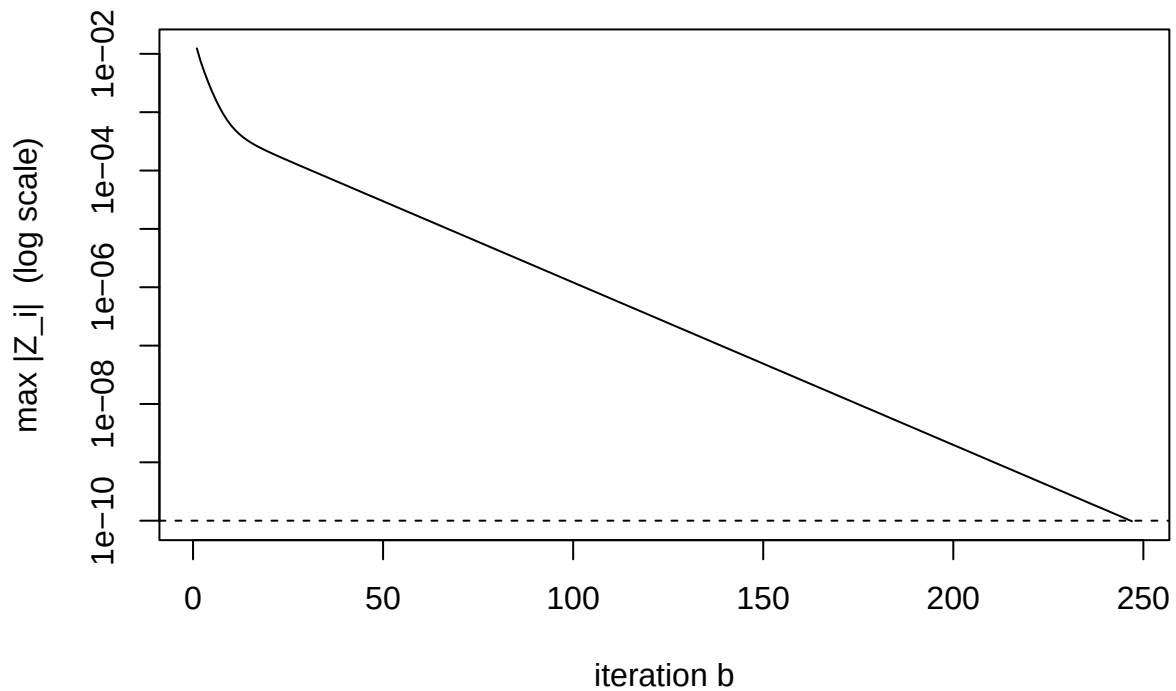
## [1] 0.012416770 0.007436664 0.004824783 0.003253452 0.002264340 0.001623858
tail(err_path2)

## [1] 1.347487e-10 1.263822e-10 1.185363e-10 1.111762e-10 1.042735e-10
## [6] 9.779999e-11

plot(err_path2, type = "l", log = "y",
     xlab = "iteration b",
     ylab = "max |Z_i| (log scale)",
     main = "Convergence with the Chinese wage numeraire")
abline(h = tol, lty = 2)

```

Convergence with the Chinese wage numeraire



Real variables

```
Phihat2 <- as.vector(pi0 %*% (That * what2^(-theta)))
Phat2    <- Phihat2^(-1 / theta)
names(Phat2) <- countries

h_ratio2 <- what2["CHN"]
Xhat2    <- y0 * what2 * Lhat + (1 - y0) * h_ratio2 * Dhat

results2 <- data.frame(
  country   = countries,
  what2     = what2,
  Phat2     = Phat2,
  real_wage2 = what2 / Phat2,
  real_exp2  = Xhat2 / Phat2
)

results2[order(-results2$real_wage2), ]
```

##	country	what2	Phat2	real_wage2	real_exp2
##	CHN	CHN 1.0207594	0.9969024	1.0239312	1.0239312
##	ROW	ROW 1.0000556	0.9998356	1.0002200	1.0011936
##	GRC	GRC 1.0001302	0.9999770	1.0001531	1.0011054
##	BGR	BGR 0.9995779	0.9995001	1.0000779	1.0005578
##	TWN	TWN 0.9996286	0.9995583	1.0000703	0.9986558
##	JPN	JPN 0.9998636	0.9997959	1.0000677	1.0002393

## AUS	AUS	0.9998694	0.9998029	1.0000665	1.0000792
## PRT	PRT	0.9996527	0.9995883	1.0000643	1.0003976
## USA	USA	1.0000000	0.9999410	1.0000590	1.0003761
## LVA	LVA	0.9993389	0.9992808	1.0000581	1.0003362
## BRA	BRA	1.0000795	1.0000244	1.0000551	1.0002897
## MEX	MEX	0.9997403	0.9996983	1.0000420	0.9999755
## GBR	GBR	0.9994647	0.9994239	1.0000407	1.0001150
## HRV	HRV	0.9993385	0.9992986	1.0000400	0.9999658
## IDN	IDN	0.9996314	0.9995933	1.0000380	0.9999441
## KOR	KOR	0.9994445	0.9994088	1.0000358	0.9991537
## FRA	FRA	0.9994001	0.9993667	1.0000334	1.0001042
## CAN	CAN	0.9997378	0.9997049	1.0000329	0.9999102
## FIN	FIN	0.9991751	0.9991442	1.0000309	0.9999033
## EST	EST	0.9991410	0.9991111	1.0000299	0.9999659
## IND	IND	0.9997589	0.9997320	1.0000269	1.0000414
## ROU	ROU	0.9992650	0.9992469	1.0000181	0.9999982
## ESP	ESP	0.9993833	0.9993679	1.0000154	0.9999584
## TUR	TUR	0.9992544	0.9992455	1.0000089	0.9997222
## LTU	LTU	0.9992114	0.9992141	0.9999973	0.9994113
## RUS	RUS	0.9989123	0.9989176	0.9999947	0.9992253
## AUT	AUT	0.9989963	0.9990111	0.9999852	0.9995387
## ITA	ITA	0.9990798	0.9990955	0.9999843	0.9996112
## POL	POL	0.9989084	0.9989261	0.9999822	0.9995981
## SVN	SVN	0.9990463	0.9990697	0.9999766	0.9994644
## SWE	SWE	0.9989493	0.9989741	0.9999752	0.9992769
## DNK	DNK	0.9990364	0.9990666	0.9999697	0.9989810
## HUN	HUN	0.9989745	0.9990089	0.9999655	0.9990927
## BEL	BEL	0.9991831	0.9992235	0.9999595	0.9992277
## SVK	SVK	0.9989926	0.9990378	0.9999547	0.9992581
## CZE	CZE	0.9988078	0.9988578	0.9999500	0.9990508
## DEU	DEU	0.9988330	0.9988834	0.9999495	0.9986667
## CHE	CHE	0.9988637	0.9989315	0.9999321	0.9984804
## NOR	NOR	0.9985504	0.9986540	0.9998962	0.9981072
## NLD	NLD	0.9987958	0.9989052	0.9998905	0.9981303
## IRL	IRL	0.9992711	0.9994234	0.9998476	0.9974293

Comparison with item 1.iii

```
compare <- data.frame(
  country      = countries,
  y0           = y0,
  real_wage_US = results$real_wage,
  real_wage_CHN = results2$real_wage2,
  diff_wage    = results2$real_wage2 - results$real_wage,
  real_exp_US  = results$real_exp,
  real_exp_CHN = results2$real_exp2,
  diff_exp     = results2$real_exp2 - results$real_exp
)

compare[order(-abs(compare$diff_wage)), ]
```

##	country	y0	real_wage_US	real_wage_CHN	diff_wage	real_exp_US
## IRL	IRL	1.1124734	1.000029	0.9998476	-1.817810e-04	1.000042
## NLD	NLD	1.0800536	1.000054	0.9998905	-1.639583e-04	1.000060

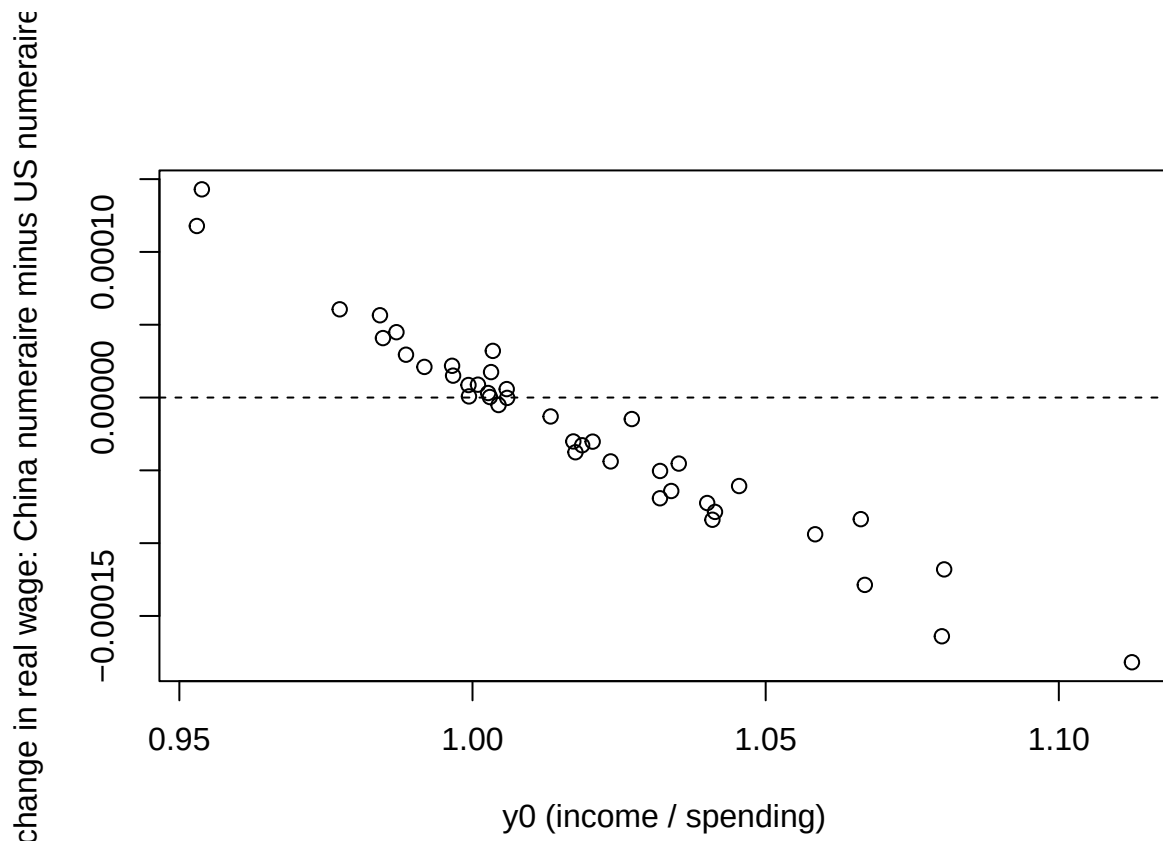
##	GRC	GRC	0.9538387	1.000010	1.0001531	1.429796e-04	1.000008
##	TWN	TWN	1.0669121	1.000199	1.0000703	-1.286510e-04	1.000238
##	NOR	NOR	1.0804442	1.000014	0.9998962	-1.180121e-04	1.000023
##	ROW	ROW	0.9529827	1.000102	1.0002200	1.178235e-04	1.000094
##	DEU	DEU	1.0584401	1.000043	0.9999495	-9.389657e-05	1.000053
##	CZE	CZE	1.0409188	1.000034	0.9999500	-8.391082e-05	1.000038
##	CHE	CHE	1.0662274	1.000016	0.9999321	-8.352362e-05	1.000024
##	KOR	KOR	1.0413576	1.000114	1.0000358	-7.853168e-05	1.000130
##	HUN	HUN	1.0400262	1.000038	0.9999655	-7.242946e-05	1.000042
##	SVK	SVK	1.0319728	1.000024	0.9999547	-6.918146e-05	1.000028
##	BEL	BEL	1.0338948	1.000024	0.9999595	-6.416889e-05	1.000027
##	DNK	DNK	1.0454711	1.000031	0.9999697	-6.076706e-05	1.000037
##	BGR	BGR	0.9773511	1.000017	1.0000779	6.059999e-05	1.000015
##	PRT	PRT	0.9842179	1.000008	1.0000643	5.651092e-05	1.000007
##	SWE	SWE	1.0319851	1.000026	0.9999752	-5.042733e-05	1.000029
##	RUS	RUS	1.0351818	1.000040	0.9999947	-4.535966e-05	1.000042
##	LVA	LVA	0.9870279	1.000013	1.0000581	4.488386e-05	1.000013
##	SVN	SVN	1.0235638	1.000020	0.9999766	-4.385891e-05	1.000022
##	USA	USA	0.9847246	1.000018	1.0000590	4.083495e-05	1.000018
##	POL	POL	1.0175580	1.000020	0.9999822	-3.756655e-05	1.000021
##	CHN	CHN	1.0186963	1.023964	1.0239312	-3.277188e-05	1.024372
##	HRV	HRV	1.0034629	1.000008	1.0000400	3.203379e-05	1.000008
##	AUT	AUT	1.0204947	1.000015	0.9999852	-3.027644e-05	1.000018
##	ITA	ITA	1.0171978	1.000014	0.9999843	-3.014168e-05	1.000016
##	BRA	BRA	0.9886575	1.000026	1.0000551	2.942820e-05	1.000024
##	GBR	GBR	0.9965166	1.000019	1.0000407	2.181393e-05	1.000019
##	JPN	JPN	0.9917930	1.000047	1.0000677	2.103252e-05	1.000045
##	MEX	MEX	1.0031638	1.000024	1.0000420	1.747824e-05	1.000024
##	FRA	FRA	0.9966876	1.000018	1.0000334	1.505396e-05	1.000018
##	LTU	LTU	1.0271723	1.000012	0.9999973	-1.477307e-05	1.000014
##	TUR	TUR	1.0133227	1.000022	1.0000089	-1.298856e-05	1.000022
##	ROU	ROU	1.0009249	1.000009	1.0000181	8.834903e-06	1.000009
##	IND	IND	0.9993101	1.000018	1.0000269	8.593701e-06	1.000018
##	CAN	CAN	1.0058330	1.000027	1.0000329	5.798769e-06	1.000027
##	IDN	IDN	1.0044427	1.000043	1.0000380	-5.131607e-06	1.000044
##	ESP	ESP	1.0026627	1.000012	1.0000154	3.102481e-06	1.000012
##	AUS	AUS	0.9993927	1.000066	1.0000665	7.881373e-07	1.000066
##	EST	EST	1.0029594	1.000030	1.0000299	3.617834e-07	1.000030
##	FIN	FIN	1.0059101	1.000031	1.0000309	-2.512028e-07	1.000032
##		real_exp_CHN		diff_exp			
##	IRL		0.9974293	-2.612433e-03			
##	NLD		0.9981303	-1.930153e-03			
##	GRC		1.0011054	1.097520e-03			
##	TWN		0.9986558	-1.581833e-03			
##	NOR		0.9981072	-1.916245e-03			
##	ROW		1.0011936	1.099180e-03			
##	DEU		0.9986667	-1.386127e-03			
##	CZE		0.9990508	-9.868106e-04			
##	CHE		0.9984804	-1.543711e-03			
##	KOR		0.9991537	-9.758808e-04			
##	HUN		0.9990927	-9.493758e-04			
##	SVK		0.9992581	-7.702802e-04			
##	BEL		0.9992277	-7.995382e-04			
##	DNK		0.9989810	-1.055828e-03			

```
## BGR      1.0005578  5.428846e-04
## PRT      1.0003976  3.910689e-04
## SWE      0.9992769 -7.520122e-04
## RUS      0.9992253 -8.170315e-04
## LVA      1.0003362  3.236643e-04
## SVN      0.9994644 -5.578122e-04
## USA      1.0003761  3.579624e-04
## POL      0.9995981 -4.228368e-04
## CHN      1.0239312 -4.407024e-04
## HRV      0.9999658 -4.246532e-05
## AUT      0.9995387 -4.795917e-04
## ITA      0.9996112 -4.049653e-04
## BRA      1.0002897  2.659368e-04
## GBR      1.0001150  9.631545e-05
## JPN      1.0002393  1.937991e-04
## MEX      0.9999755 -4.894421e-05
## FRA      1.0001042  8.618611e-05
## LTU      0.9994113 -6.026031e-04
## TUR      0.9997222 -3.001045e-04
## ROU      0.9999982 -1.113637e-05
## IND      1.0000414  2.312671e-05
## CAN      0.9999102 -1.168778e-04
## IDN      0.9999441 -9.955297e-05
## ESP      0.9999584 -5.403094e-05
## AUS      1.0000792  1.369267e-05
## EST      0.9999659 -6.383361e-05
## FIN      0.9999033 -1.285292e-04
```

```
# the difference lines up with the sign of the trade imbalance
cor(compare$diff_wage, compare$y0)
```

```
## [1] -0.9752906
```

```
plot(compare$y0, compare$diff_wage,
      xlab = "y0 (income / spending)",
      ylab = "change in real wage: China numeraire minus US numeraire")
abline(h = 0, lty = 2)
```



Switching the numeraire changes almost nothing for China but reshuffles everyone else along a single dimension. The correlation between the change in the real wage and y_n^0 is -0.975 . The reason is that only one term in the system differs: $(1 - y_n^0) \hat{w}_{\text{China}} \hat{D}_n$. Since $\hat{w}_{\text{China}} > 1$, every transfer is now worth more in real terms. Deficit countries have $1 - y_n^0 > 0$ and receive that transfer, so they spend more and their wage rises. Surplus countries pay it and lose.

Ireland and the Netherlands run the largest surpluses in the sample ($y^0 = 1.11$ and 1.08) and lose the most. Greece and the rest-of-world aggregate run the largest deficits and gain the most. The effects on real expenditure are roughly ten times larger than on real wages, up to 0.26 percentage points for Ireland, because the transfer enters spending directly and reaches wages only through general equilibrium.

The magnitudes are small here because 2014 imbalances are modest, with y^0 ranging from 0.95 to 1.11. The point is that the choice of $h(w)$ is not a harmless normalisation. It fixes the real value of international transfers and therefore redistributes between debtors and creditors.